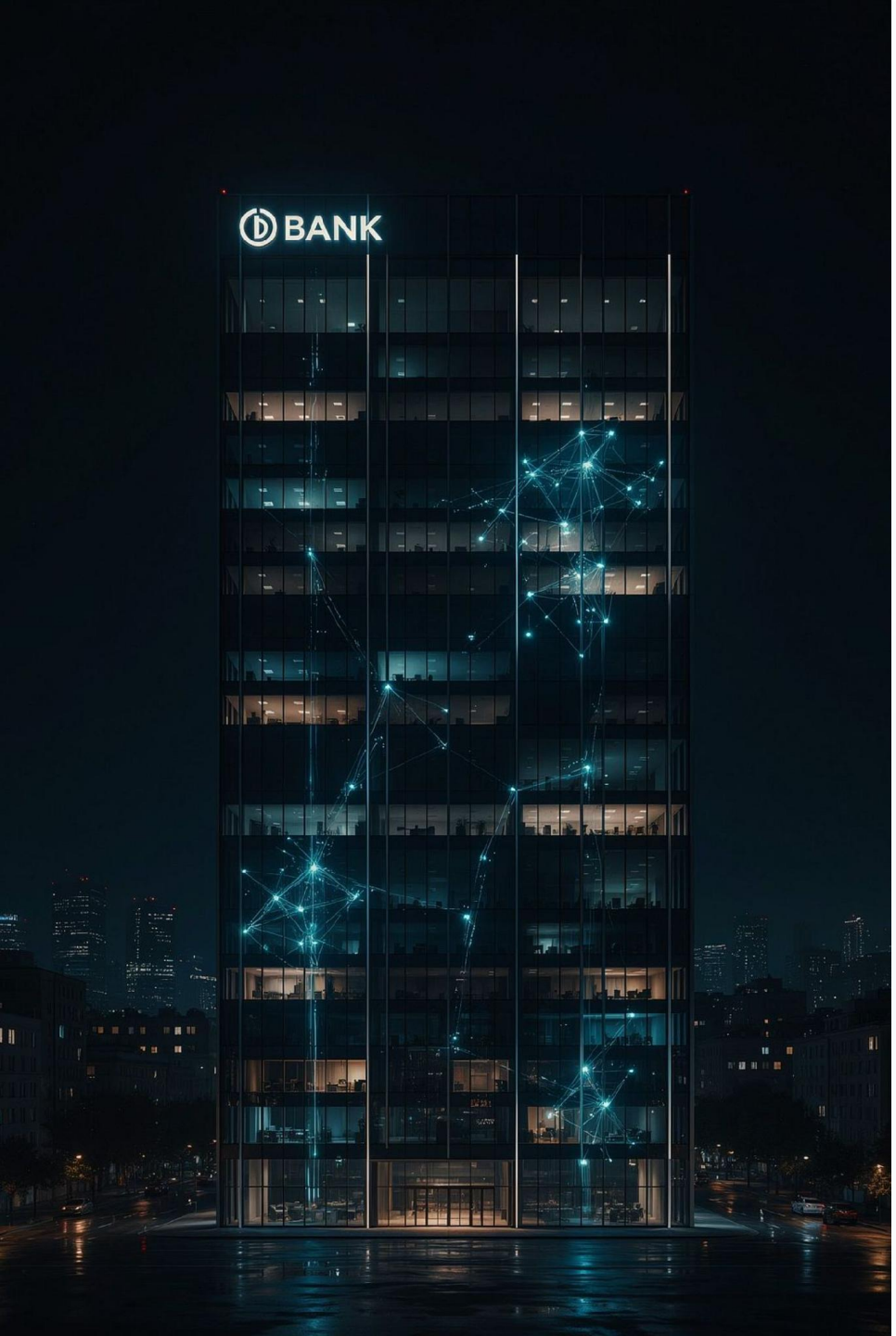


Prompt Engineering for AML: Building Reliable Compliance Workflows at Scale

Engineering Trustworthy AI for Financial Crime Compliance and Regulatory Reporting

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TRANSFORMING COMPLIANCE OPERATIONS THROUGH GROUNDED, AUDITABLE, AND SCALABLE AI WORKFLOWS

About the Speaker

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14+ years delivering technology transformation across financial services, AI, and regulatory compliance. Supported TD Bank, RBC, Santander, and United Missouri Bank across AML, fraud detection, risk management, and cloud initiatives.

AWS Certified Solutions Architect · MS Computer Science, NYIT

AI & GenAI for Banking

Enterprise LLM strategy and agent-based automation

AML & Regulatory Compliance

Financial crime workflows and reporting frameworks

Cloud-Native Architecture

Scalable enterprise modernization on cloud platforms

Fraud & Risk Analytics

AI-driven detection models and risk management



The Growing Complexity of AML Compliance

Financial institutions process millions of transactions daily while navigating evolving regulatory expectations and resource-intensive investigation requirements. Traditional workflows struggle to keep pace.

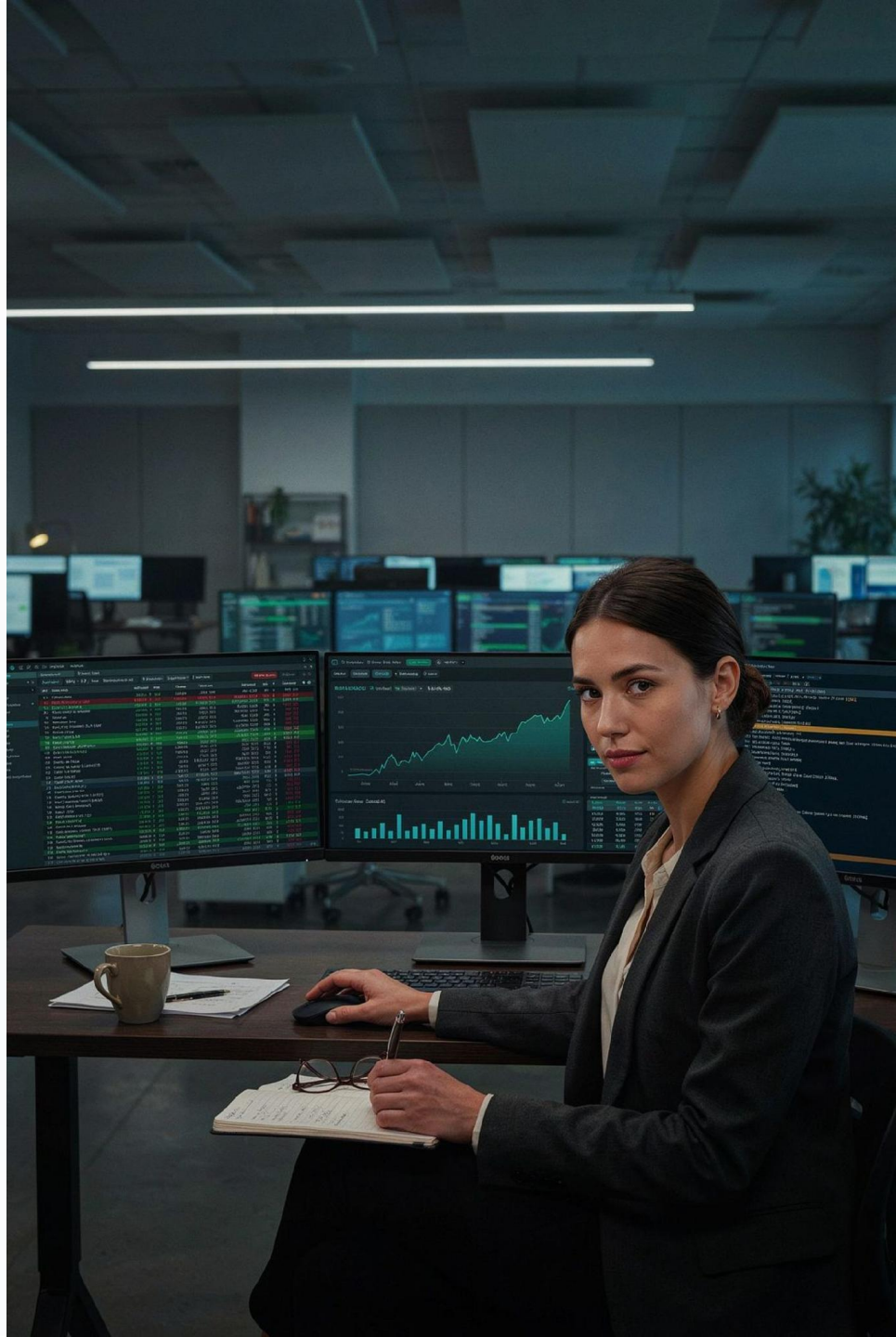
Key Operational Challenges

- High alert volumes requiring investigation and documentation
- Fragmented customer and transaction data across siloed systems
- Inconsistent documentation and limited decision traceability
- Manual narrative preparation for regulatory reporting

The AI Opportunity

Large language models can help analysts organize evidence, identify information gaps, and draft investigation narratives — dramatically reducing preparation time while maintaining compliance rigor.

AI should enhance compliance workflows without replacing accountable human judgment.



What Prompt Engineering Means in AML

Beyond simple instructions — prompts are compliance controls. In AML, a poorly designed prompt can produce plausible but unsupported conclusions with serious regulatory consequences.

Five Essential Design Principles

1 Context Grounding

Use only approved transaction records, customer data, and verified investigation evidence

2 Structured Instructions

Define clear tasks, permitted interpretations, and output requirements

3 Evidence Traceability

Connect every generated statement to verifiable source record IDs

4 Controlled Reasoning

Distinguish observed facts from hypotheses and investigative conclusions

5 Safe Failure

Flag missing or conflicting evidence rather than inventing explanations

Prompt Design: Basic vs. Governed

❌ Basic Prompt

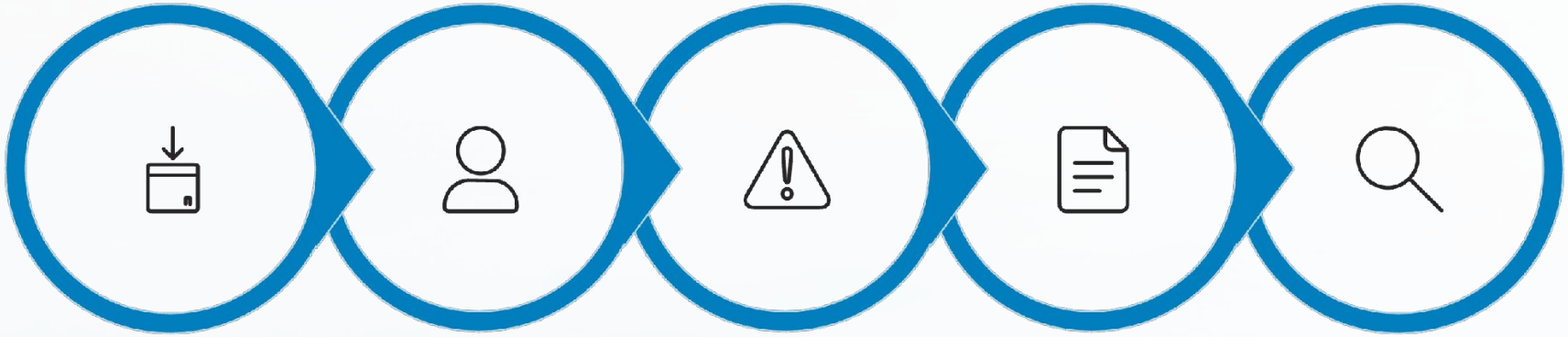
"Explain why this transaction is suspicious."

✅ Governed Prompt

"Summarize the supplied transaction evidence, identify documented alert indicators, cite supporting record IDs, and flag information gaps. Do not infer criminal intent."

AML and Currency Transaction Reporting Workflows

AI-assisted AML workflows must integrate seamlessly with transaction monitoring systems and regulatory reporting obligations across jurisdictions.



Ingest Enrich Alert Summarize Investigate

AML Investigations

Assess potential suspicious activity based on behavioral patterns and documented evidence

Currency Transaction Reporting

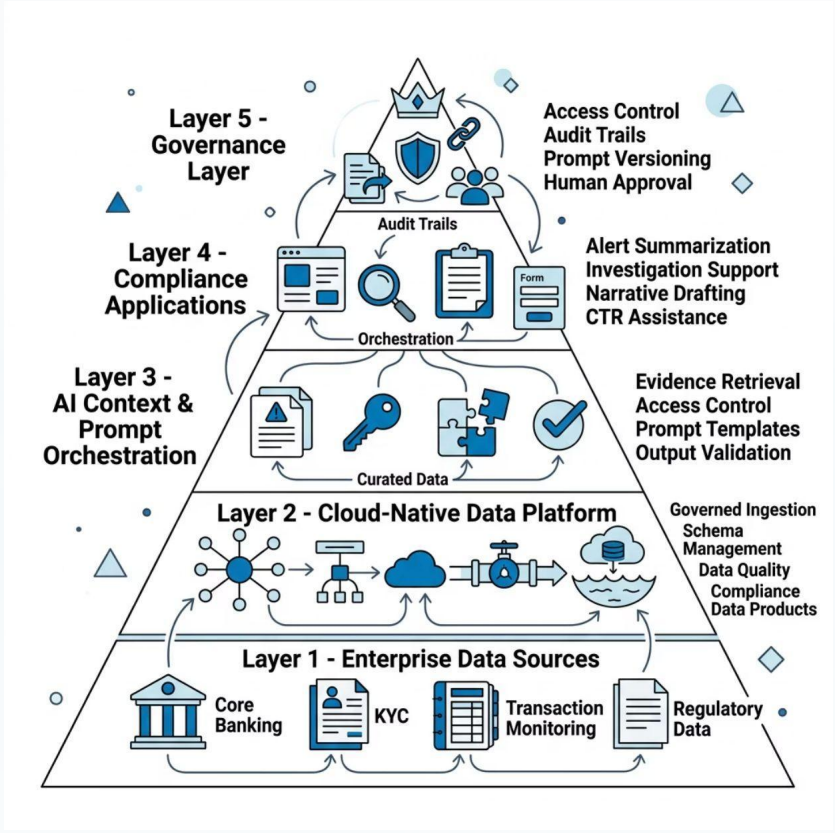
Follows jurisdiction-specific thresholds and mandatory reporting requirements

Human Authority

AI supports evidence preparation; compliance personnel retain full validation and submission authority

Reference Architecture for AI-Assisted AML

A governed, layered architecture ensures that AI applications connect to trusted enterprise data while enforcing access control, auditability, and human oversight at every layer.



Architecture Design Principles

Enterprise Data Sources

Core banking, payment systems, KYC records, and compliance reference data

Cloud-Native Data Platform

Governed ingestion pipelines with quality controls and curated compliance data products

AI Orchestration Layer

Approved retrieval, context filtering, reusable prompt templates, and model validation

Compliance Applications

Alert summarization, narrative drafting, and CTR preparation assistance

Governance Layer

Access control, audit trails, monitoring, prompt versioning, and human approval gates

COMPLIANCE | RETRIEVAL | TRACEABILITY

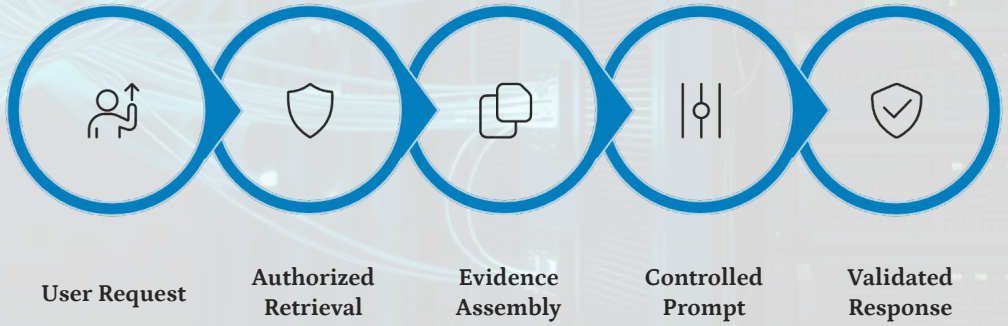
Grounding AI Responses in Trusted Financial Data

LLMs can produce fluent but unsupported statements when relevant transaction evidence is absent. Reliable outputs begin with reliable, authorized context.

Grounding Strategy Requirements

- Retrieve only from **authorized compliance datasets** with role-based access restrictions
- Include transaction timestamps, amounts, account relationships, and **evidence identifiers**
- Distinguish **verified records** from analyst notes and unverified third-party inputs
- Carry **evidence references** into all generated outputs for downstream traceability

RAG Pattern for Compliance



Every material factual claim in an investigation narrative must be traceable to a supporting evidence record.

Designing Structured Prompts for AML Investigations

A production-grade AML prompt is not a single instruction it is a governed template with clearly separated components, each serving a specific compliance function.

Prompt Component Framework

Component	Purpose
Role	Define the AI as an investigation-support assistant
Task	Specify the required investigation activity precisely
Context	Supply approved transaction and case evidence only
Constraints	Prohibit unsupported assumptions and inferences
Output Schema	Enforce predictable, machine-readable results
Validation	Define evidence and completeness requirements

Illustrative Prompt Template

```
ROLE: AML investigation support assistant.

TASK: Summarize supplied transaction activity.

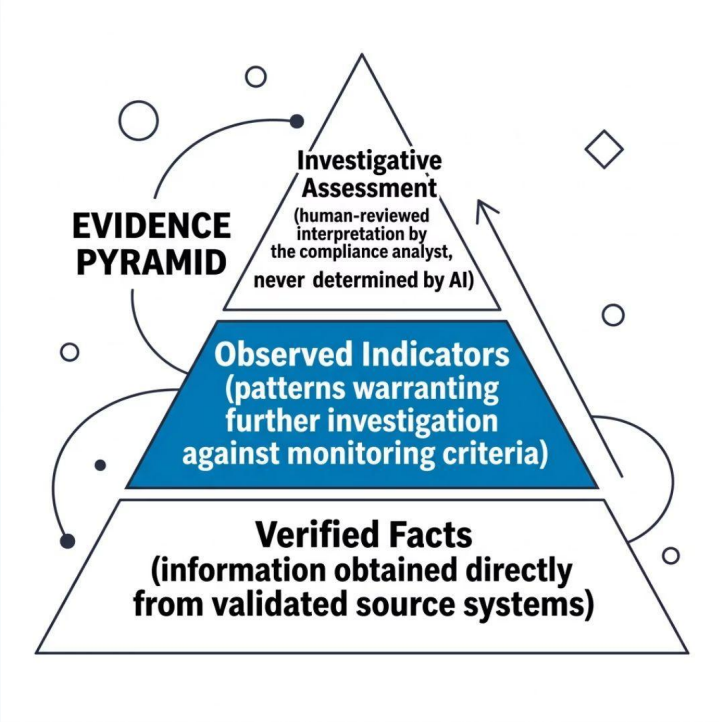
CONTEXT: Use only authorized case evidence.

RULES:
- Do not invent transactions or customer details
- Do not infer criminal intent
- Cite evidence IDs for factual statements
- Flag missing or conflicting information

OUTPUT:
- Transaction summary
- Observed risk indicators
- Evidence references
- Missing information
- Analyst review requirements
```

Separating Facts, Interpretation, and Decisions

Maintaining clear boundaries between transaction facts and investigative interpretation is a critical requirement in financial crime compliance — and a core responsibility of prompt design.



1

Verified Facts

Information obtained directly from validated source systems.
"The supplied records show multiple transfers occurring during the same reporting period."

2

Observed Indicators

Patterns that may warrant further review against monitoring criteria.
"Transaction timing and frequency warrant review against the institution's thresholds."

3

Investigative Assessment

An interpretation that must be reviewed and supported by the compliance analyst — never determined autonomously by AI.

⚠ AI-generated indicators are not proof of wrongdoing. The model must never independently determine regulatory filing outcomes.

Schema-Aware Prompt Engineering and Validation

Unstructured AI responses create dangerous integration gaps in downstream compliance platforms. Enforcing structured, schema-validated outputs is a production requirement, not an optimization.

Structured Output Example

```
{
  "case_id": "AML-CASE-001",
  "transaction_summary": "Activity summary",
  "observed_indicators": [
    {
      "indicator": "Transaction pattern",
      "evidence_ids": ["TXN-001", "TXN-002"]
    }
  ],
  "missing_information": [],
  "review_status": "PENDING_ANALYST_REVIEW"
}
```

Validation Gate Requirements



Schema Enforcement

Enforce JSON schema and expected data types; reject malformed responses



Evidence Verification

Verify all identifiers against retrieved case evidence; reject unreferenced claims



Completeness Check

Detect missing mandatory fields and invalid status values



Human Escalation

Route failed validation to correction workflow or analyst review



A valid JSON response is not necessarily a factually correct compliance assessment.

Building Safe and Reliable AI Workflows

Reliable AML systems must respond safely when evidence is incomplete, contradictory, or unavailable. Fail-safe design is a first-class engineering requirement.

Failure Scenarios and Controls

Failure Scenario	Required Control
Missing transaction evidence	Request additional evidence
Conflicting customer records	Flag inconsistency for investigation
Unsupported model claims	Reject or escalate the output
Unauthorized data access	Block retrieval and generation
Invalid output format	Validate and retry within limits
Low evidence confidence	Require mandatory human review

Additional Safeguards

> In regulated operations, a safe refusal or escalation is always better than a confident, unsupported answer.

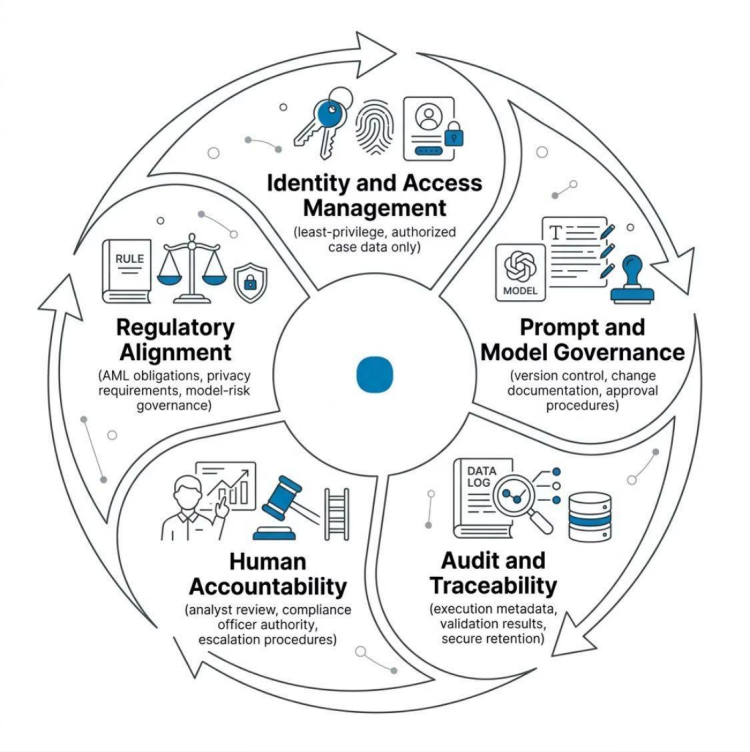
> Treat retrieved documents as **untrusted content**, not executable instructions — apply prompt-injection defenses

> **Prohibit autonomous regulatory submission** under all circumstances

> Escalate repeated validation failures and preserve intervention records

Governance, Auditability, and Human Oversight

Financial crime compliance demands transparent, controlled, and reviewable AI-assisted processes. Governance is not a post-launch concern — it is a foundational engineering requirement.



Identity & Access

Enforce least-privilege; restrict model access to authorized case data only

Prompt Governance

Version-controlled prompts with documented configurations and production approval gates

Audit & Traceability

Retain execution metadata, validation results, and review actions per regulatory retention policy

Human Accountability

Analysts review all AI summaries; compliance officers retain full regulatory decision authority

Practical Implementation Roadmap

From pilot prompt to enterprise AML capability a phased approach that builds governance, confidence, and operational maturity at each stage before scaling.

Phase 1

Identify Use Cases
Prioritize bounded, human-reviewed tasks: transaction summarization and evidence organization

Phase 2

Governed Data
Connect authorized transaction datasets, customer context, and compliance references

Phase 3

Engineer Prompts
Develop templates, output schemas, evidence controls, and safe-failure rules

Phase 4

Validate & Pilot
Security testing, compliance review, regression testing, and supervised analyst trials

Phase 5

Operationalize
Monitoring, change management, analyst feedback loops, and continuous evaluation

Consistent Documentation

Greater analyst documentation uniformity

Evidence Traceability

Improved linkage from outputs to source records

Reduced Manual Work

Fewer hours on avoidable preparation tasks

Operational Visibility

Stronger monitoring and controlled review processes

Key Takeaways and Closing

1

Prompt Engineering Is an Engineering Discipline

Reliability requires structured instructions, controlled context, and repeatable validation not intuition

2

Grounding Creates Defensibility

Material factual claims must be linked to authorized transaction and investigation evidence

3

Schema and Validation Are Essential

Structured outputs enable safer, auditable integration with downstream compliance workflows

4

Governance Must Be Built In

Security, observability, auditability, and prompt versioning are core production requirements

5

Human Judgment Remains Central

AI supports investigation workflows while accountable professionals retain full decision authority

> "Trustworthy AI in financial crime compliance is not defined by how convincingly a model responds, but by how reliably its outputs can be verified, governed, and defended."

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